

[1] Index

Rev	Date	Designer	Description
A	25.01.26	Ganghyeok Lim	Create design project

Index

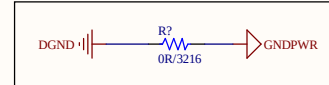
#1 Index	#8 ??
#2 Overview	#9 ??
#3 ??	
#4 ??	
#5 ??	
#6 ??	
#7 ??	

Title Sitara_Project_Rev_A.PrjPcb		Rev A
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Sheet # 1 of 10	Engineer Ganghyeok Lim	
Date 2025-01-26		
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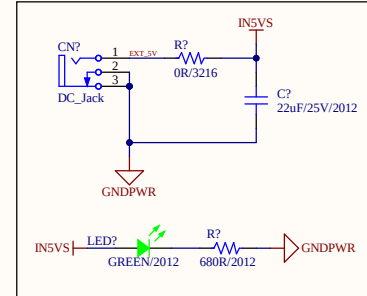
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A	Sitara Project B'd (Rev.A) [2] Overview																															
B																																
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D	<table><tr><td colspan="3">Title</td><td>Rev</td></tr><tr><td colspan="3">Sitara_Project_Rev_A.PrjPcb</td><td>A</td></tr><tr><td colspan="2">Doc</td><td colspan="2" rowspan="3"></td></tr><tr><td colspan="2">Overview.SchDoc</td></tr><tr><td>Sheet # 2 of 10</td><td>Engineer</td></tr><tr><td colspan="2"></td><td colspan="2">Ganghyeok Lim</td></tr><tr><td colspan="2">Date</td><td colspan="2">2025-01-26</td></tr><tr><td colspan="4">This document is copyright of ArkX and shall not be revealed, produced, copied, in whole or in part, nor used for any purpose other than submitted.</td></tr></table>				Title			Rev	Sitara_Project_Rev_A.PrjPcb			A	Doc				Overview.SchDoc		Sheet # 2 of 10	Engineer			Ganghyeok Lim		Date		2025-01-26		This document is copyright of ArkX and shall not be revealed, produced, copied, in whole or in part, nor used for any purpose other than submitted.			
Title			Rev																													
Sitara_Project_Rev_A.PrjPcb			A																													
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	1	2	3	4																												

[3] Power

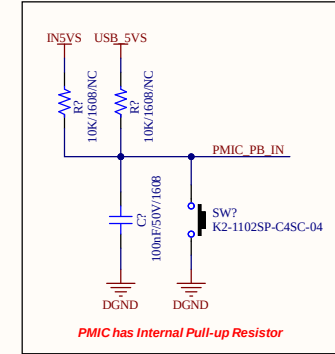
Digital GND



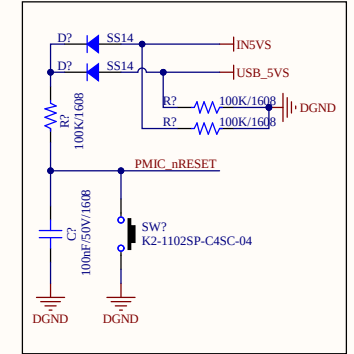
Power Connector (5V)



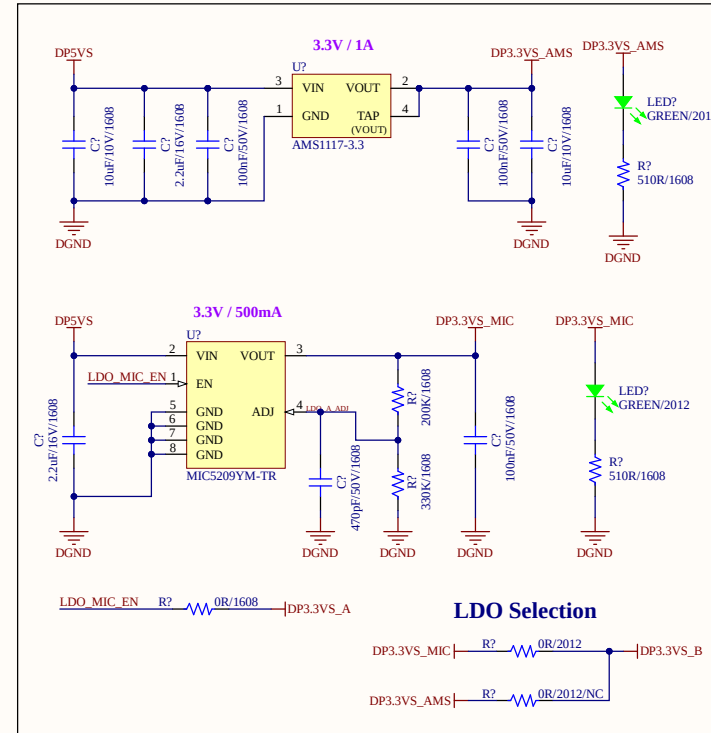
PMIC Power-On Button



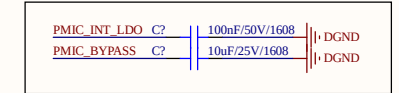
PMIC Reset Button



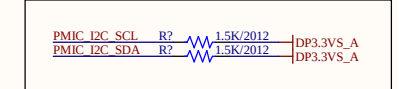
LDO (3.3V / 500mA)



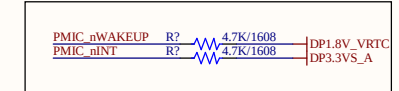
INT LDO / BYPASS Pin



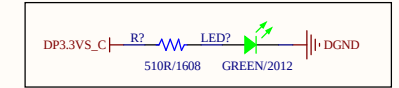
I2C Pull-up Resistor



nWAKE-UP / nINT Pull-up



PMIC Power LED



LDO Selection

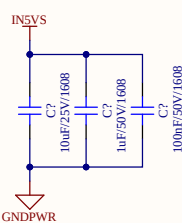


PMIC's Output Voltage based on AM3358's OPP100

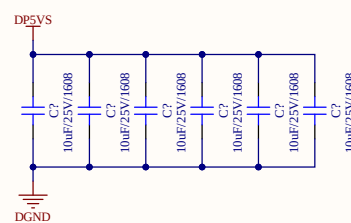
- DCDC1 (1.5V / 1.2A) => CPU's "VDDS_DDR"
- DCDC2 (1.1V / 1.2A) => CPU's "VDDS_MPU"
- DCDC3 (1.1V / 1.2A) => CPU's "VDDS_CORE"
- LDO1 (1.8V / 0.1A) => CPU's "VIO" & "VDDS" & "VDDS_RTC"
- LDO2 (3.3V / 0.1A) => No Connection with CPU
- LDO3 (1.8V / 0.4A) => Some Connection with CPU
- LDO4 (3.3V / 0.4A) => Some Connection with CPU

Decoupling Capacitors

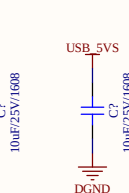
Decap for IN5VS



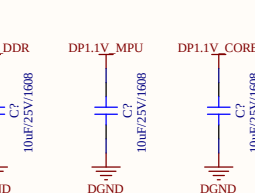
Decap for DP5VS



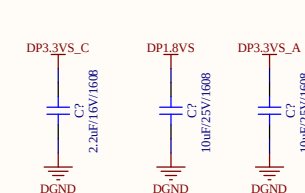
Decap for USB_5VS



Decap for DCDC 1/2/3

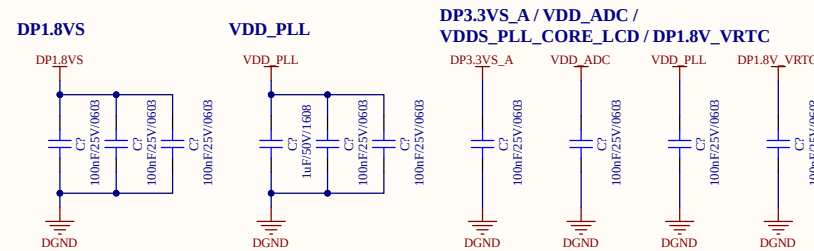
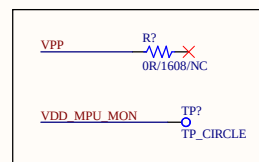
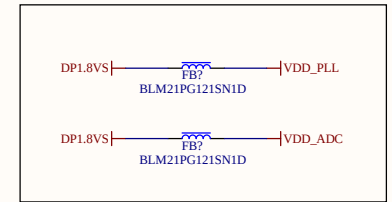
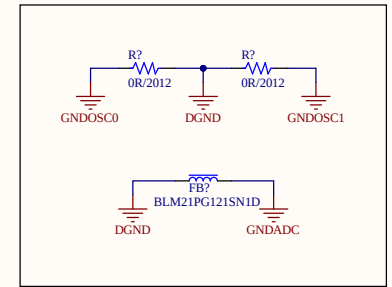
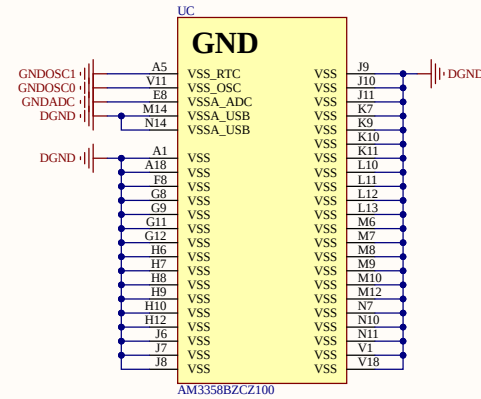
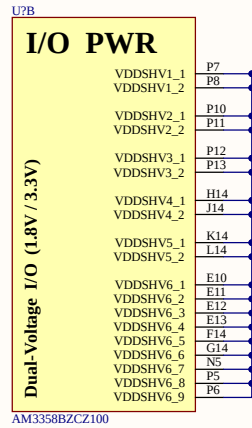


Decap for LDO 1~4



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Place Decaps Closely to Each Pin of PMIC



CAP_VDD_RTC	C?	1uF/50V/1608	DGND
CAP_VBB_MPU	C?	1uF/50V/1608	DGND
CAP_VDD_SRAM_MPU	C?	1uF/50V/1608	DGND
CAP_VDD_SRAM_CORE	C?	1uF/50V/1608	DGND

DP1.1V_MPU

DP1.8V_VDDS

DP3.3VS_A

DP1.1V_CORE

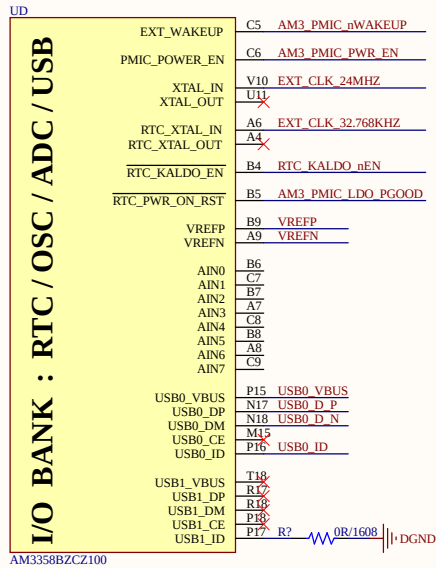
DP1.5V_DDR

VDDSHV1~6

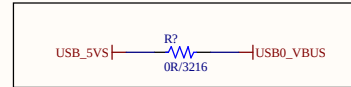
DP3.3VS_A => VDDA3P3V_USB0/I (100nF)
 DP1.8VS => VDDA1P8V_USB0/I, VDDS_SRAM_CORE_BG, VDDS_SRAM_MPU_BB (100nF)
 VDD_PLL => VDDS_OSC(1uF), VDDS_PLL_MPU, VDDS_PLL_DDR (100nF)

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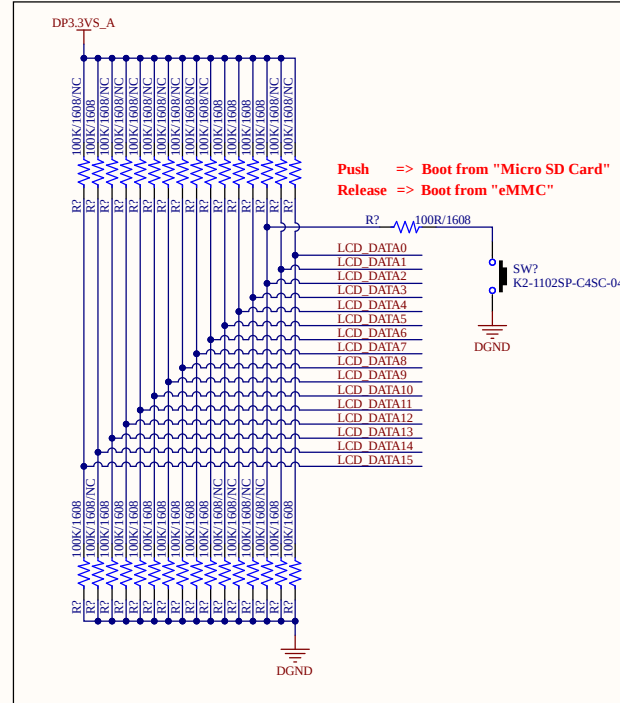
[5] Sitara AUX Part



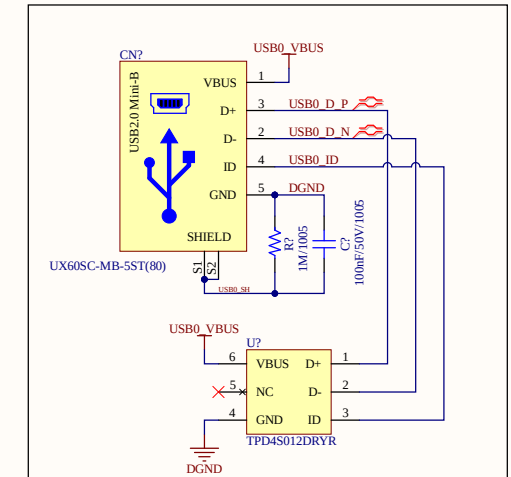
USB Power



Boot Config



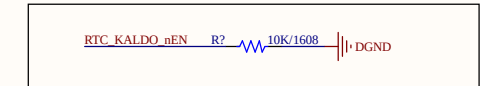
USB 2.0 Mini-B Connector



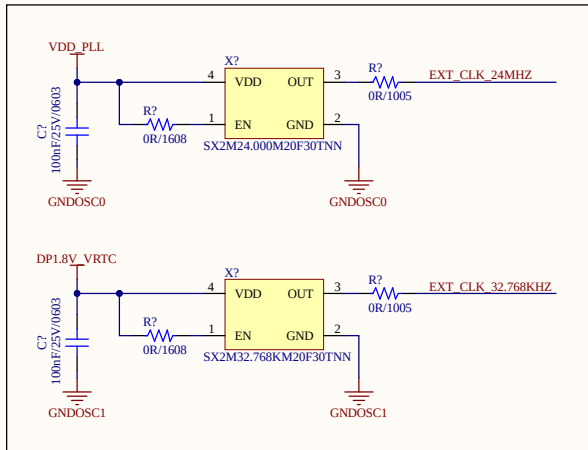
Indication LED (USB0 VBUS)



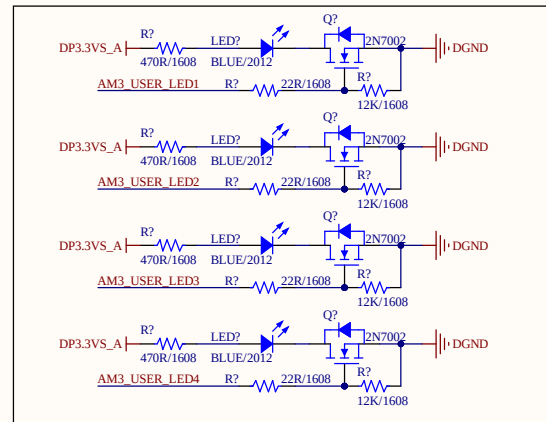
Internal RTC LDO Enable



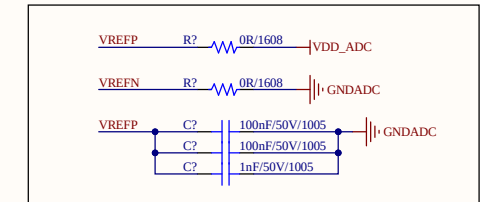
External Oscillator (24MHz / 32.768kHz)



User LEDs

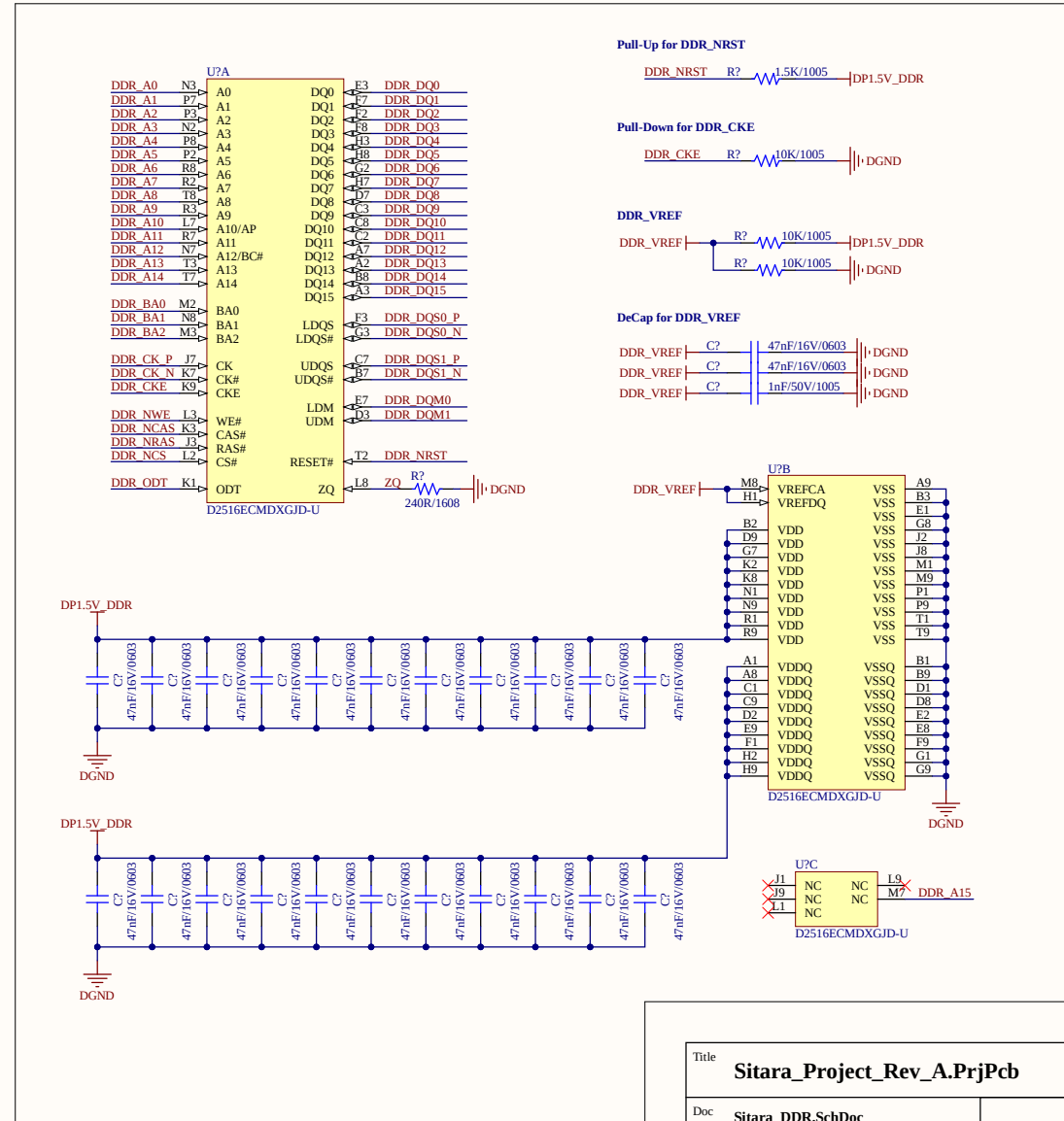


Analog Reference



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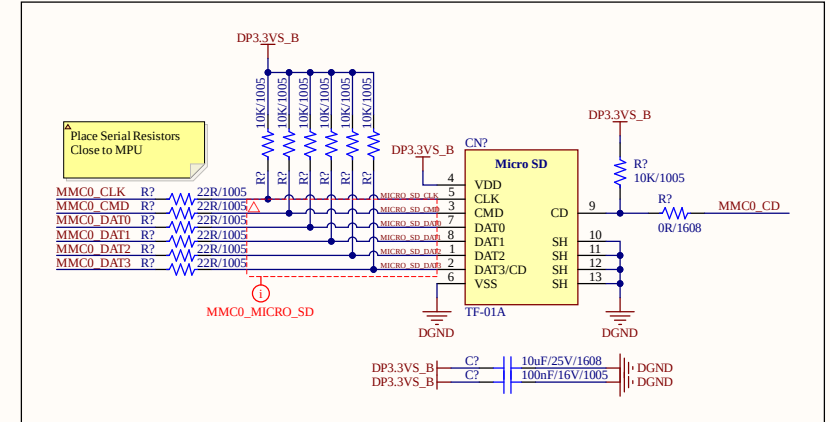
DDR3L Memory



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MICRO SD Card (MMC0)

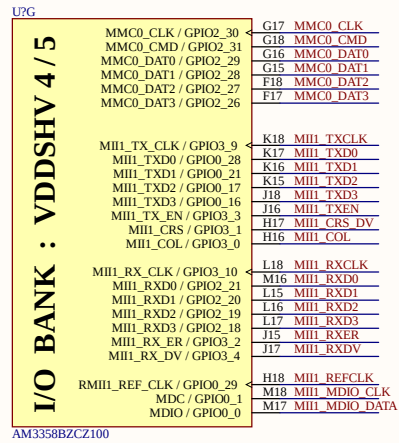
The diagram shows an AT24C16 EEPROM chip connected to a DP3.3VS_A supply and an I2C bus. The chip's VCC pin (pin 1) is connected to the supply. The SCL pin (pin 2) is connected to the I2C bus. The SDA pin (pin 3) is connected to the I2C bus. The GND pin (pin 4) is connected to ground. The WP pin (pin 5) is connected to a 4.7k pull-up resistor to the supply. The NC pins (pins 6, 7, and 8) are also connected to 4.7k pull-up resistors to the supply. The chip is labeled 'AT24C16' and 'I2C ADDR: 1010_XXXX'. The bus lines are labeled 'AM3_I2C0_SCL', 'AM3_I2C0_SDA', and 'AM3_EEPROM_WP'.



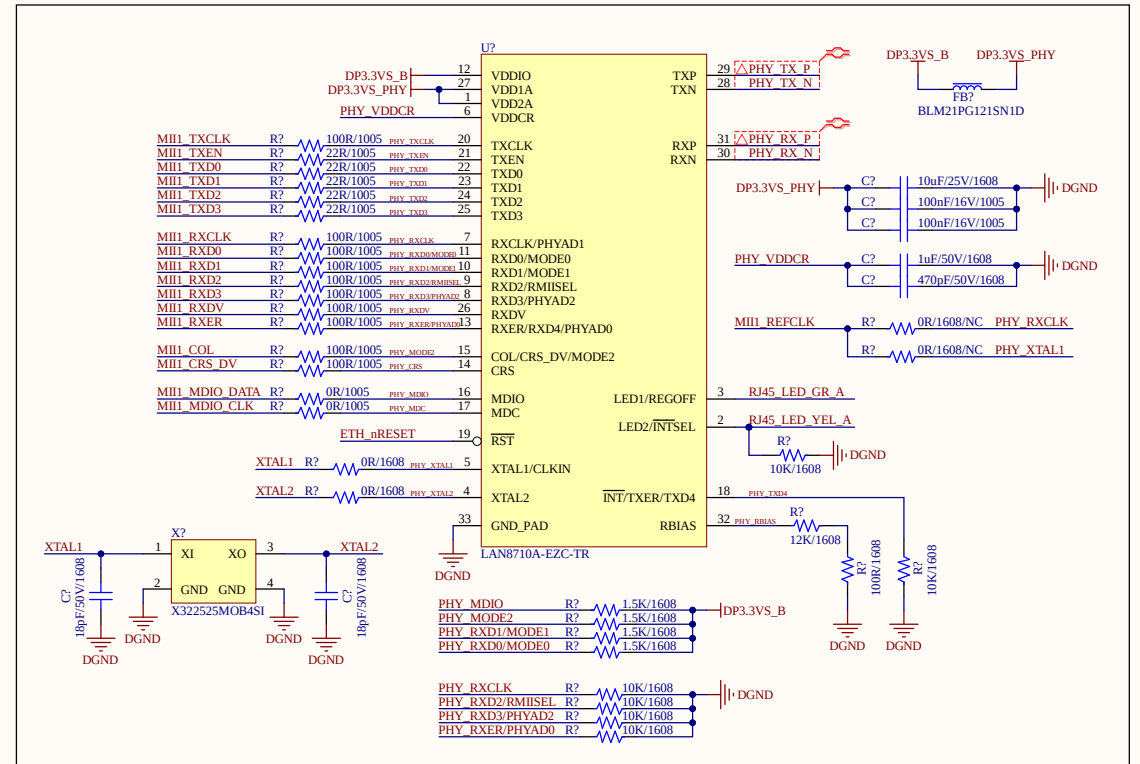
K7	RF14	NC61	J1
K6	RF15	RF14	H4
K5	RF12	NC59	H3
K4	RF11	NC58	H2
K3	VSF7	NC57	H1
K2	VSF6	NC55	H0
K1	VSF5	NC54	G4
J10	VSF3	NC53	G3
J9	VSF2	NC51	G2
J8	VSF1	NC50	G1
J7	NC109	NC49	F4
J6	NC108	NC48	F3
J5	NC107	NC46	F2
J4	NC105	NC44	F1
J3	NC104	NC43	F0
J2	NC103	NC42	E4
J1	NC102	NC41	E3
I14	NC101	NC40	E2
I13	NC100	NC39	E1
I12	NC99	NC38	E0
I11	NC97	NC36	D4
I10	NC96	NC35	D3
I9	NC95	NC34	D2
I8	NC94	NC33	D1
I7	NC93	NC32	D0
I6	NC91	NC31	C4
I5	NC90	NC29	C3
I4	NC89	NC28	C2
I3	NC88	NC26	C1
I2	NC87	NC25	C0
I1	NC86	NC24	B4
H10	NC85	NC24	B3
H9	NC84	NC23	B2
H8	NC83	NC22	B1
H7	NC82	NC21	B0
H6	NC81	NC20	A4
H5	NC80	NC19	A3
H4	NC79	NC18	A2
H3	NC78	NC17	A1
H2	NC77	NC16	A0
H1	NC76	NC15	A0
G10	NC75	NC14	A0
G9	NC74	NC13	A0
G8	NC73	NC11	A0
G7	NC72	NC10	A4
G6	NC71	NC9	A3
G5	NC70	NC8	A2
G4	NC69	NC7	A1
G3	NC68	NC6	A0
G2	NC66	NC5	A0
G1	NC65	NC4	A3
F10	NC64	NC3	A3
F9	NC63	NC2	A1

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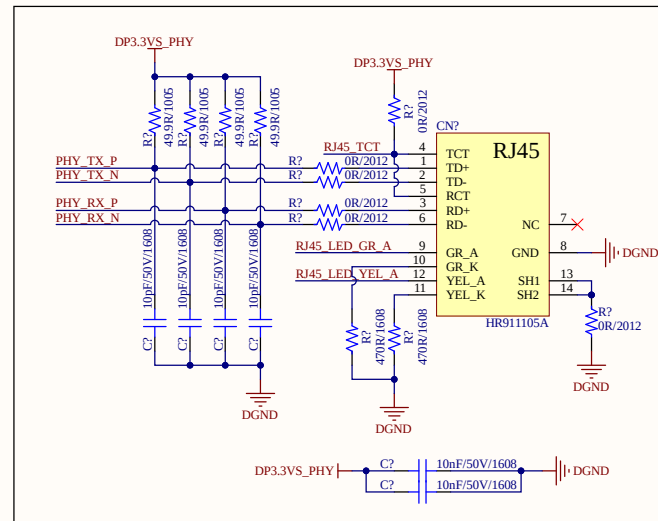
[9] Sitara I/O Bank 4/5



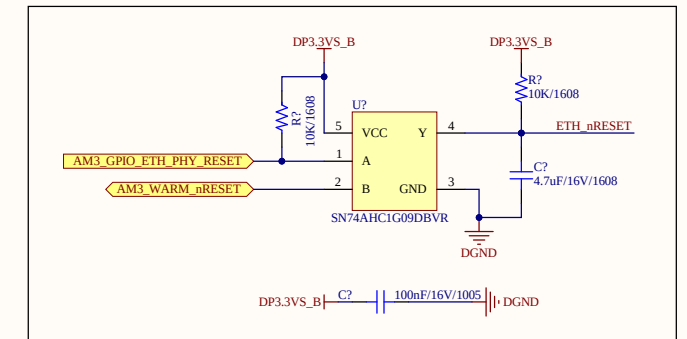
Ethernet PHY (10/100Mbps)



RJ45 Connector

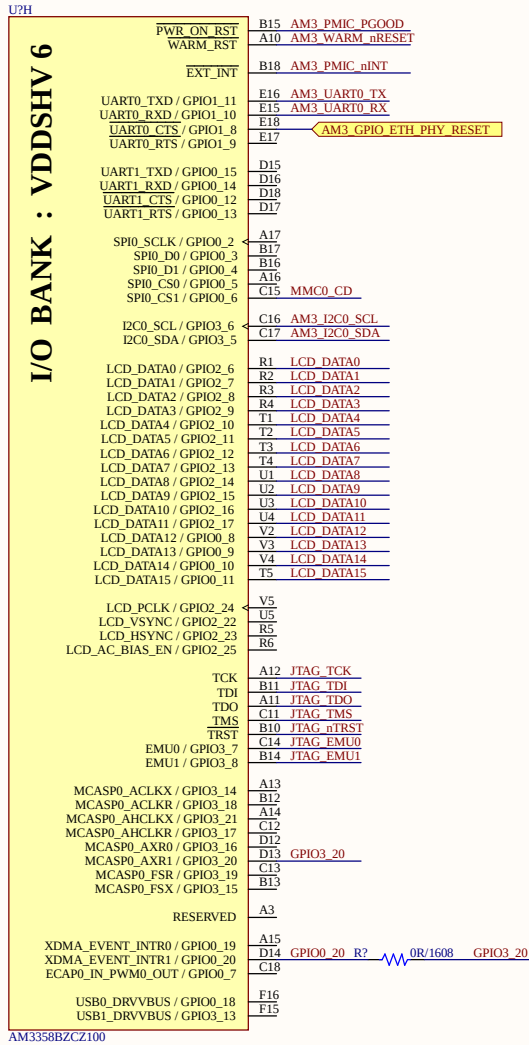


Ethernet PHY Reset

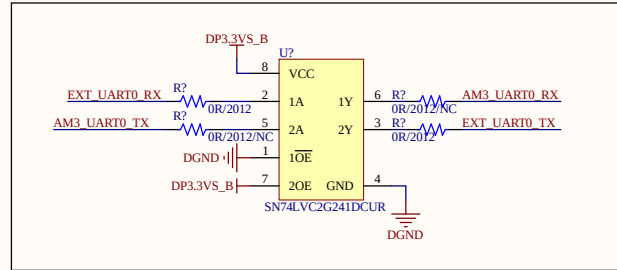


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[9] Sitara I/O Bank 6

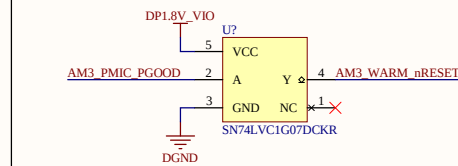


UART0 External Connector

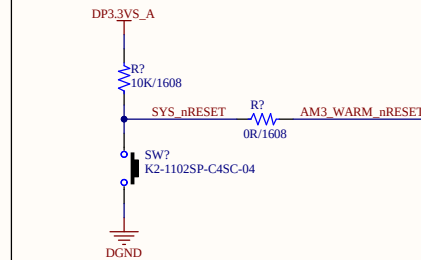


System Reset

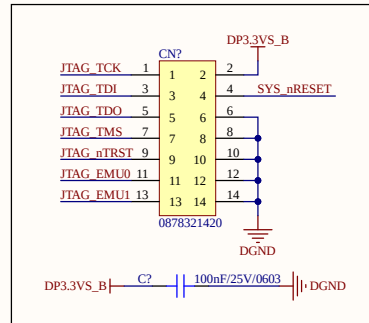
Power-On Reset



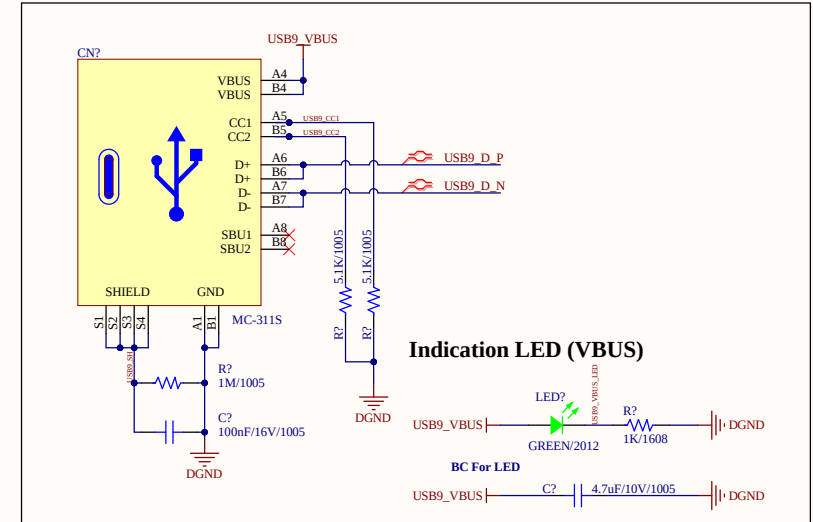
Reset Switch



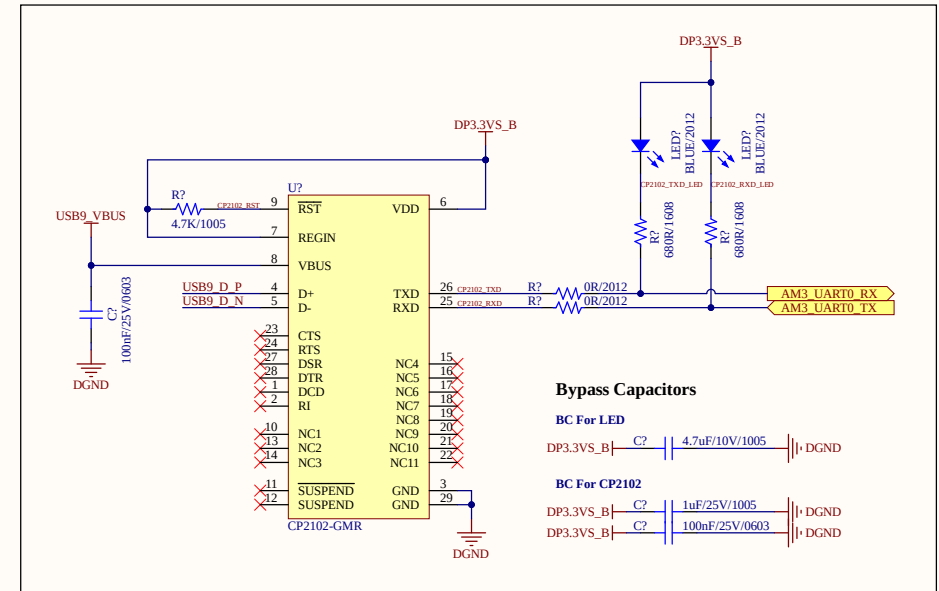
JTAG Connector



USB Connector (Type-C)



USB - UART Bridge



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[?] Schematic Design Check-List

Schematic Design Check-List

No.	Category	Item	Description	Confirm